VINEETH KALLURU

Applied AI Engineer | Production LLM Systems, Agents and Post-Training | IEEE Senior Member

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SUMMARY

Applied AI engineer who takes open models and agents from POC to production inside customer environments. At NVIDIA, embedded with 25+ Tier-1 enterprise software and industrial customers plus Microsoft Azure and OpenAI: model selection and evals, post-training data, inference sizing and tuning, agent harnesses, and guardrails. Diagnoses customer-blocking failures, ships the fix, and feeds it back to the product teams. Earlier work on TPU performance modeling and a dataflow AI compiler. Co-author of NVIDIA Nemotron Nano 2; author of the Nemotron-CC data pipeline; first author on accepted agentic-AI papers at IEEE ICA 2026 and ISSRE 2026.

EXPERIENCE

AI Engineer, Rudhira Biosciences (RudhiraBio) **2026 - Present**

- Building an AI-driven neoantigen discovery platform for Lynch syndrome cancers on AWS HealthOmics: bulk RNA-seq and WES pipeline with AI-based neoantigen ranking.
- Architecting the customer-facing product for biopharma computational teams and clinician-researchers: tiered service outputs, multi-institutional data intake, cohort-level MSI stratification.

Senior Solutions Architect, Generative AI, NVIDIA (Strategic ISV AI Foundry, Santa Clara, CA) **Nov 2023 - 2026**

- **POC to production at scale:** Technical owner of GenAI architecture and adoption for 25+ Tier-1 enterprise software and industrial customers (data management, MLOps, semiconductor manufacturing, logistics, financial services) and hyperscale partners Microsoft Azure and OpenAI; owned engagements from discovery and evals through production and executive reviews.
- **Zoom:** Co-led the Zoom x NVIDIA on-prem AI Companion announcement at GTC 2026 with Dell and CDW. Diagnosed customer-blocking failure modes in a document-parsing model, built a decoding guard that Zoom adopted, and fed the root cause back to the model team; sized on-prem GPU deployments and led embedding and guardrail evaluation.
- **Cohesity:** Owned the Gaia architecture featured in the GTC 2026 keynote: model evaluation and recommendation, inference and GPU sizing, guardrails, and executive-escalation support. Handed the account to a successor in Oct 2026 and remain its SME. Recognized with an NVIDIA NIM Coin nomination for the engagement.
- **Glean:** Originated the partnership that put Nemotron 3 Nano behind Glean's production LLM router (GA Apr 2026). Wrote the serving requirements behind a merged NVIDIA Dynamo PR, ran a citation-failure root-cause study that fed Nemotron post-training, and negotiated NVIDIA's first Evaluation Data License Agreement with a strategic ISV.
- **Inference at scale:** Architected multi-node and disaggregated LLM serving on H100, H200, GB200 NVL72 and GB300 NVL72 with TensorRT-LLM, NVIDIA Dynamo, NIM and Run:ai; defined on-prem, hybrid and air-gapped reference architectures. NVIDIA technical lead for Azure's MLPerf Inference v4.0 submission (46% gain on LLaMA-2 70B; Azure blog, HPCWire).
- **Post-training, evals and agent harnesses:** Led Nemotron fine-tuning workshops and post-training data design across partners (including a domain-specific procurement model with NeMo Curator and Data Designer). Built reusable eval harnesses for RAG, document parsing and agent tool-calling that partners adopted for their own methodology. Built NVIDIA's multi-agent Asset Lifecycle Management reference (NeMo Agent Toolkit).
- **Open source:** Authored the Nemotron-CC pretraining data pipeline in NeMo Curator (6.7K-line PR), adopted by NAVER HyperCLOVA X, Amazon Rufus and Moonshot AI.
- **Field to roadmap:** Filed roughly 30 field bugs and RFEs and maintained the consolidated enterprise feedback on Nemotron, NeMo Retriever, NeMo Guardrails, NV Ingest and NIM; several shipped as fixes or process changes.
- **Multiplier:** Mentored four SAs and an intern on customer engagements, delegated a flagship account with a documented handoff, and served on SA hiring panels. Won a GTC 2026 Spot Award for the DLI workshop below.

ML Accelerator Architect, Cloud TPU, Google **Jul 2022 - Nov 2023**

- Performance analysis of Gemini workloads on TPU v6 and v7 compute and memory subsystems; hardware-software co-design under Norm Jouppi and David Patterson.

Compiler / ML Systems Engineer (Team Lead), SambaNova Systems **Feb 2020 - Jul 2022**

- Led a 7-engineer team that shipped the first known production for-loop and RNN support in a dataflow compiler: about 4x faster online inference and 5x faster LSTM training vs. NVIDIA A100 (SambaNova public blog, Aug 2022); about 3x chip-utilization gain from BERT tiling and compiler work.

SELECTED PROJECTS

- **FinanceHarness:** Finance retrieval harness for search agents powered by sub-10B models; accepted at the RL4LLM-Agents workshop, ACM ICAIF 2026. Companion: **rl-for-ai-research**, GRPO training on FinanceBench (open source).
- **WG-Plan:** Workflow-grounded small-model planners for domain-specific tool calling, using workflow graphs to constrain valid actions (research in progress).
- **jarvis-desk-bot:** Hermes agent voice front end on an ESP32 desk robot (local speech, spoken approvals); MIT licensed.

SELECTED PUBLICATIONS

- **Between Weights and Workflows: Executable Graph Contracts for Bounded Enterprise Agents.** IEEE ICA 2026, Kumamoto. Accepted (oral); first author.
- **The Coherence Trap: Layer-Decomposed Failure-Mode Discovery in a Deployed Agentic LLM System.** IEEE ISSRE 2026 Industry Track (CORE A). Presented; first author.
- **The Agentic Data Platform: A Contention Model and Reference Architecture for Multi-Agent Workloads on the Lakehouse.** IEEE BigData 2026 (19.2% acceptance). Accepted; co-author.
- **Beyond the Blue Link** (GEO in stochastic retrieval) and **Edit-Distance Links** (conditional-tokenization world models). ACM Hypertext 2026, published; third author on both.
- **SkillOpt:** Trajectory-Derived, Verifier-Grounded Compilation of LLM-Agent Skills. ACM CAIS 2026 Agent Skills Workshop; 63% fewer tokens and 62% lower latency at non-regressing reward. **Provenance Is the Control Plane** (AI4DEMONS, CIKM 2026, accepted); **Sovereign AI Networking** (AAAI HCOMP 2026 workshop).
- **NVIDIA Nemotron Nano 2** (arXiv:2508.14444, co-author); **Llama-Embed-Nemotron-8B** (arXiv:2511.07025, contributor); **HANNA / FbNet** (arXiv:1906.07214, first author; IEEE ICCSP 2018). Two granted Indian patents in applied AI/ML (IN 431037, IN 451829).

SPEAKING

- NVIDIA GTC 2026 full-day DLI workshop (co-instructor) and poster; GTC DC 2025; Microsoft Build 2024 and Ignite 2024; SC25 demo; SEMI Agentic AI for Semiconductor Manufacturing 2025.
- O'Reilly AI Superstream 2026 ("Building Agent Harnesses to Match Frontier-Model Performance in RAG"); AIKP 2026 invited workshop; TechEquity AI Summit 2026 (Glean demo, Nov 5); CMU guest lecturer (18-740).

RECOGNITION AND SERVICE

- **IEEE Senior Member**; IEEE-HKN; NVIDIA NIM Coin nomination and GTC 2026 Spot Award; winner, Apple-sponsored Memory Fairness Protocol competition (CMU).
- **Reviewer / PC:** IEEE TKDE; PC member, IEEE ICA 2026, KDD 2026 AgenticSE and Agent4IR; reviewer, NeurIPS 2026 (ethics and workshops), ACM CAIS 2026, KDD 2026 workshops. IEEE Computer Society Elevation Committee.
- **Judge:** NVIDIA Generative AI Agents and LlamaIndex developer contests; Harness Engineering Hack (2026). **Technical reviewer:** AI Agents: The Definitive Guide and Hands-On LLM Serving and Optimization (O'Reilly).

SKILLS

Inference: TensorRT-LLM, NVIDIA Dynamo, NIM, vLLM, Run:ai, disaggregated and multi-node serving, benchmarking (MLPerf)
Models and agents: Nemotron and open LLMs, post-training data design, GRPO, RAG, agent harnesses, NeMo Curator, NeMo Guardrails, NeMo Agent Toolkit, LLM evaluation
Systems and cloud: Python, C/C++, CUDA, ML compilers, Kubernetes; Azure (AzureML, AI Foundry, AKS), AWS (HealthOmics), GCP; on-prem and air-gapped

EDUCATION

M.S., Electrical and Computer Engineering, Carnegie Mellon University — **Aug 2018 - Dec 2019**

Professional Program in Artificial Intelligence, Stanford University — **Sep 2022 - Dec 2022**

B.Tech., Electronics and Communication Engineering, SRM Institute of Science and Technology — **Jul 2014 - Apr 2018**